

ANSWER KEY — Integration Lesson 24 Test: Tangent, Secant & Friends

For the grader · full working shown · deduct only half for a missing + C · sign errors on the "co" functions lose the point

1. Exact values of $\tan(\pi/3)$ and $\sec(\pi/3)$.

Work: $\tan(\pi/3) = \sin(\pi/3)/\cos(\pi/3) = (\sqrt{3}/2)/(1/2) = \sqrt{3}$. $\sec(\pi/3) = 1/\cos(\pi/3) = 1/(1/2) = 2$.

Answer: $\tan(\pi/3) = \sqrt{3}$, $\sec(\pi/3) = 2$.

2. $\tan 0$, $\sec 0$, $\tan(\pi/4)$; where is $\tan x$ undefined and why?

Work: $\tan 0 = 0/1 = 0$; $\sec 0 = 1/1 = 1$; $\tan(\pi/4) = (\sqrt{2}/2)/(\sqrt{2}/2) = 1$. Undefined where the bottom is zero: $\cos(\pi/2) = 0$, so $\tan(\pi/2)$ is undefined (any odd multiple of $\pi/2$ accepted). **Answer:** 0, 1, 1; $\tan$ is undefined at $x = \pi/2$ because $\cos(\pi/2) = 0$ (division by zero).

3. $\int \sec^2 x \, dx$ with check.

Work: reading $(\tan x)' = \sec^2 x$ backward: $\int \sec^2 x \, dx = \tan x + C$. Check: $(\tan x)' = \sec^2 x$ ✓.

Answer: $\tan x + C$. (The check must appear — it was asked for. Common wrong answer: $\sec^3 x/3 + C$, the power-rule-on-trig trap, gets 0.)

4. $\int \csc x \cot x \, dx$.

Work: $(\csc x)' = -\csc x \cot x$, so we need the minus out front: $\int \csc x \cot x \, dx = -\csc x + C$. Check: $(-\csc x)' = +\csc x \cot x$ ✓. **Answer:** $-\csc x + C$. (+ $\csc x + C$ is the expected sign slip — no credit, the sign IS the content here.)

5. $\int (2\sec^2 x + \cos x) \, dx$.

Work: split and use the library: $\int 2\sec^2 x \, dx = 2 \tan x$; $\int \cos x \, dx = \sin x$. Check: $(2 \tan x + \sin x)' = 2 \sec^2 x + \cos x$ ✓. **Answer:** $2 \tan x + \sin x + C$.

6. $\int_0^{\pi/4} \sec^2 x \, dx$.

Work: FTC: $\tan x \big|_0^{\pi/4} = \tan(\pi/4) - \tan 0 = 1 - 0 = 1$. **Answer:** 1.

7. $\int_0^{\pi/3} \sec x \tan x \, dx$.

Work: antiderivative is $\sec x$. FTC: $\sec(\pi/3) - \sec 0 = 2 - 1 = 1$. **Answer:** 1. ($\sec(\pi/3) = 1/\cos(\pi/3) = 1/(1/2) = 2$; $\sec 0 = 1$.)

8. $\int \sec^2(3x) \, dx$ with check.

Work: linear-inside rule: base antiderivative $\tan x$, inside $3x$, so divide by 3: $\tan(3x)/3 + C$. Check: derivative of $\tan(3x)/3 = (1/3) \cdot \sec^2(3x) \cdot 3 = \sec^2(3x)$ ✓. **Answer: $\tan(3x)/3 + C$.** ($\tan(3x) + C$ without the $1/3$ is the standard chain-rule miss — no credit; the check was required and would have caught it.)

9. Student claims $\int \csc^2 x \, dx = \cot x + C$. Right or wrong?

Work: differentiate the claim: $(\cot x)' = -\csc^2 x \neq +\csc^2 x$ — wrong by a sign. Correct: $\int \csc^2 x \, dx = -\cot x + C$, since $(-\cot x)' = \csc^2 x$ ✓. **Answer: wrong; the correct antiderivative is $-\cot x + C$.** (Full credit needs the differentiation check shown, the verdict, and the fix.)

10. $\int \tan x \, dx$ by u-substitution, every step.

Work: $\int \tan x \, dx = \int (\sin x / \cos x) \, dx$. Let $u = \cos x \rightarrow du = -\sin x \, dx \rightarrow \sin x \, dx = -du$. Integral becomes $\int (-du)/u = -\int (1/u) \, du = -\ln|u| + C = -\ln|\cos x| + C$. Check: derivative = $-(1/\cos x)(-\sin x) = \tan x$ ✓. **Answer: $-\ln|\cos x| + C$ (equivalently $\ln|\sec x| + C$ — either form earns full credit).** (Missing absolute-value bars: half credit; missing the back-substitution to x : half credit.)